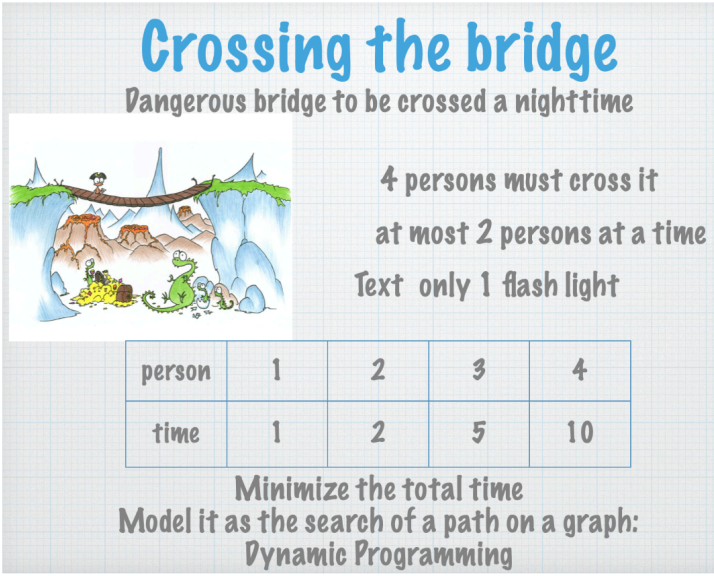
Homework 5



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To model the problem as the search of a path on a graph, we can think of it as the **shortest path problem** and describe the objective function as the lowest possible cost obtainable by a path from *start* to *finish*. We can then use the following formulation:

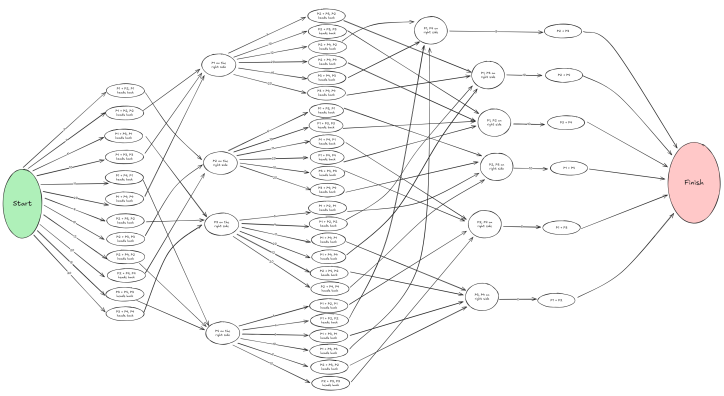
$$\text{Sets: A, N}$$

$$\text{Parameters: } c_{ij} \text{ costs, } \forall (ij) \in A \text{ and start, finish } \in N$$

$$\text{Variables: } x_{i,j} = \begin{cases} 1 & \text{if (i,j)} \in \text{path} \\ 0 & \text{else} \end{cases}$$

$$\text{Constraints: } \forall i \in N, \sum_{(sj) \in FS(s)} x_{sj} = 1 \text{ and } \sum_{(it) \in BS(t)} x_{it} = 1$$

$$\text{Objective Function: } min \sum_{(ij) \in A} c_{ij} \cdot x_{ij}$$



BridgeFSM.png

The modeled graph where we can apply our reasoning takes into consideration all of the possible paths and **assuming to not include useless moves** (i.e: a person goes back at the start of the bridge with another one).